Vocabulary Expansion on SmolLM2-135M

Objective

Adapt a small pretrained language model to Greek language in two ways, and measure which adaptation works better and what it costs the original model.

Time estimate is 1-2 days of work. Training is meant to be the small part. Most of the time should go towards design and analysis. Do not train to converge, that is not needed. You will be judged on the way you design the setup.

Model

Use SmolLM2-135M (https://huggingface.co/HuggingFaceTB/SmolLM2-135M). It has 135M parameters, with a 49k token tokenizer.

Dataset

Greek text, roughly 100MB. Suggested sources: Greek Wikipedia (the el config of wikimedia/wikipedia) plus a web slice from CulturaX (uonlp/CulturaX, el) or the Greek portion of OSCAR. You can pull this from any data source of your choice.

100MB is the pool, not the training target. It is enough to train a basic tokenizer, run both experiments mentioned below, and hold out clean evaluation splits with room to spare.

Task

Run two experiments, then compare them.

Experiment 1: fine-tune

Continue training SmolLM2-135M on the Greek corpus with the original tokenizer unchanged.

Experiment 2: extend the vocabulary, then fine-tune

Extend SmolLM2's vocabulary with new Greek tokens, then adapt the model on the Greek corpus. Target roughly 8,000 new vocab tokens. The how is yours to figure out. This is where you need to think about the design.

The condition that makes the comparison valid

Match the training budget across the two experiments: same tokens, same steps, same compute. Report the off-the-shelf model as the shared baseline for both. Train for a max of 30 min per experiment. The aim is to see how the 2 approaches differ and not till model convergence.

Evaluation

You build the evaluation set. Its design is part of what we assess.

Compare Experiment 1 and Experiment 2 on two axes:

1. Greek language-modeling quality on held-out Greek text.

2. English capability retained, on held-out English text. Experiment 1 is the reference: measure how much extra English Experiment 2 gives up for its Greek gain.

Compute

You should be able to run this experiment on the Google Colab free version. A free Colab T4 is enough.

Feel free to run this on any other compute resource you have access to.

Deliverables

1. Code that runs both experiments and the evaluation, as a repository or a google colab we can execute.
2. A short writeup, one to two pages: what you did, the results table, and your reading of the tradeoff. We are interested to see your learnings and how you modify the existing vocabulary

We care more about the reasoning and the experimental design more than a polished result.

Practical notes

Run sizing: keep each training run short, on the order of tens of millions of tokens, targeting around 30 min. Subsample the corpus for the runs.

Use of AI tools is encouraged. We want you to have a deep understanding of what you are building and manage your agents well!